REAL TIME BEST FRAME SELECTION AND MARKER FREE GARMENT SIZE MEASUREMENT FOR SMART APPAREL PACKING

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

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Abstract

Camera-based garment measurement requires visible boundaries, consistent measurement locations and reliable conversion from pixels to physical dimensions. This research addresses the integration of frame-suitability checks, segmentation and temporal confirmation for garment measurement within a controlled overhead-camera workflow. The aim was to develop a dimension-measurement and size-estimation component for flat T-shirts, shirts and trousers in the Garment Intelligence platform. Garment footage was acquired at a fixed camera height of 100 cm, and selected images were annotated with segmentation polygons. The final dataset contained 1,295 images, comprising 1,070 training, 141 validation and 84 test images, with one garment class. A YOLOv8n-seg (You Only Look Once version 8 nano segmentation) model was trained for 50 epochs at an image size of 640. The final logged epoch achieved validation mask precision of 0.97707, recall of 0.97037, mean average precision (mAP) at 0.50 of 0.96033 and mAP@0.50-0.95 of 0.95529. The deployed component combines garment geometry, calibrated conversion, size-chart mapping, three-frame confirmation and removal-dependent counting with a FastAPI service and SQLite storage. Five recorded T-shirt and shirt comparisons against tape measurements produced width and length mean absolute errors of 0.570 cm and 0.674 cm respectively. The first two records are repeated trials of one garment. These preliminary results support further investigation of the implemented workflow but do not establish general measurement accuracy, size agreement or counting reliability. Independent test-set evaluation, broader garment trials and operator validation remain necessary. The individual contribution comprises dataset preparation, model training and integration, geometric measurement, temporal confirmation and application integration.

Keywords: Garment measurement, Instance segmentation, Camera calibration, Temporal confirmation, Size estimation

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I thank my research group members for their collaboration in bringing our individual components together within the Garment Intelligence platform. Working together on the desktop application strengthened my understanding of component integration and the importance of clear software interfaces.

I am also grateful to those who supported my visits to garment manufacturing environments and the collection of garment video footage. These opportunities helped me understand practical measurement conditions and prepare images for the segmentation dataset.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CPU	Central Processing Unit
CSV	Comma-Separated Values
F1	F1 Score (Harmonic Mean of Precision and Recall)
GPU	Graphics Processing Unit
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
JSON	JavaScript Object Notation
LiDAR	Light Detection and Ranging
MAE	Mean Absolute Error
mAP	Mean Average Precision
PR	Precision-Recall
RMSE	Root Mean Square Error
UI	User Interface
YOLO	You Only Look Once

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1 INTRODUCTION

1.1 Overall Project Context

The Garment Intelligence platform brings production monitoring, garment rework counting, style and colour analysis, and garment size measurement into a shared desktop user interface (UI). Each research component retains its own service and responsibilities. The individual work reported in this thesis concerns the measurement component; the other components provide the wider application context.

1.2 Individual Component Context

My personal research component involves measuring the length and width of garments using camera images and estimating the garment size based on these measurements. This system has been developed for T-shirts, shirts, and trousers. In this process, the mere presence of a garment in an image is insufficient for obtaining measurements; factors such as proper garment positioning, clear visibility of the sections to be measured, and measurement stability across consecutive frames must also be verified.

The initial input for the component consists of the camera image provided by the frontend and the selected garment type. A You Only Look Once version 8 nano segmentation (YOLOv8n-seg) model is used to identify the region of the image corresponding to the garment. Subsequently, checks regarding the detection zone and garment geometry are performed, and the length and width are obtained in pixels based on measurement points specific to the selected garment type.

A fixed-height, marker-free calibration method is employed to convert pixel measurements into actual dimensions. The camera is positioned directly overhead at a height of 100 cm above the measurement surface. In the current configuration, a calibration value of 13.6 pixels per centimeter is used for an image resolution of 1920 × 1080. This scale is used to convert the pixel dimensions into centimeters. To minimize the risk of confirming a measurement based on transient movements or segmentation fluctuations, the system verifies measurement stability across three consecutive frames. Once confirmed, the measurement is compared against the size

chart relevant to the garment type to estimate the size. An annotated image displaying length, width, size, and measurement lines is then displayed on the frontend.

In addition to the core measurement process, this component tracks garment counts by size, measurement history, and production-order records. This data is stored in an SQLite database, enabling the user to review both current results and past records.

1.3 Problem Statement

The primary challenge in measuring the length and width of garments using camera images lies in identifying a suitable image frame and extracting stable, accurate dimensions from it. Even when a garment is positioned in front of the camera, the boundaries required for measurement may not be clearly identifiable due to factors such as the garment being in the process of being arranged, being folded, being obscured by hands or other objects, or having parts extending beyond the camera's field of view. Measurements derived from such a frame may not accurately represent the garment's true dimensions.

Variations in shape and the specific points requiring measurement across different garment types also complicate this issue. For instance, using the total width of a T-shirt-including the sleeves-as its "chest width" is inappropriate. Therefore, rather than simply obtaining the length and width of the bounding box surrounding the garment, it is necessary to identify specific measurement points relevant to the garment type.

Another challenge involves converting pixel dimensions from the image into actual measurements in centimeters. Since factors such as camera height, image resolution, and the specific capture setup influence this conversion, a calibration value tailored to the specific setup must be employed. Failure to do so can result in incorrect physical measurements, even if the garment segmentation itself is accurate. Variations in measurements and size estimates caused by slight movements and segmentation differences between consecutive frames must be taken into account. Furthermore, to prevent the same garment from being counted repeatedly while positioned under the camera, it is necessary to distinguish between the stages of securing, counting, and removing the garment.

Accordingly, this research addresses the challenge of identifying suitable garment frames under a controlled overhead camera setup, consistently obtaining length and width measurements specific to the garment type, converting these values into centimeters, and estimating the size based on a specific size chart. To validate the success of this method, it is necessary to evaluate the system's measurements by comparing them with actual reference measurements.

1.4 Research Gap

Research on the automated extraction of garment dimensions using computer vision technology has already been conducted. The study titled “An Intelligent Solution for Automatic Garment Measurement Using Image Recognition Technologies” presents a method for obtaining garment measurements by integrating garment segmentation with the extraction of measurement key points [1]. Another study, “Automatic Measurements of Garment Sizes Using Computer Vision Deep Learning Models and Point Cloud Data,” derives measurements by combining image-based key-point detection with light detection and ranging (LiDAR) depth and point-cloud data [2]. Therefore, garment segmentation or camera-based measurement alone cannot be considered the novel contribution of this research.

This research focuses on the practical requirement of identifying the optimal moment for measurement within a continuous camera feed to obtain stable garment dimensions. In a production environment, not all frames captured while placing, adjusting, and removing a garment under the camera are equally suitable for measurement. Consequently, beyond merely detecting the garment in a single image, it is necessary to evaluate garment visibility, segmentation geometry, and measurement consistency across consecutive frames.

Accordingly, the research gap addressed here involves integrating frame-suitability checks, garment segmentation, calibrated dimension measurement, and size estimation into a unified process—and evaluating its performance—under a fixed overhead camera setup. The term "marker-free" in this context signifies that no reference marker is required within each runtime image. This does not imply that the camera setup requires no initial calibration.

To address this requirement, segmentation datasets were prepared using selected frames from video footage collected in actual garment manufacturing environments. The developed component verifies measurement stability across three consecutive frames and compares the confirmed dimensions against the size chart relevant to the specific garment type. It also manages the processes of garment removal and preparation for the next measurement to minimize the risk of repeatedly measuring the same garment.

Highlighting this research gap is not a claim that existing methods lack these capabilities; rather, it identifies a specific requirement examined within the context of the controlled camera setup and garment workflow adopted in this study. To validate the proposed method, it is necessary to evaluate measurement error against reference measurements and assess the impact of frame-stability checks.

1.5 Aim and Objectives

1.5.1 Main objective

The main objective of this research is to develop and evaluate a component that identifies suitable garment frames under a controlled, fixed-height overhead camera setup, determines garment length and width using segmentation and calibration techniques, and estimates the size based on these dimensions.

1.5.2 Sub objectives

1.Data collection and preprocessing

Collecting video footage in actual garment manufacturing environments at a fixed camera height of 100 cm, extracting frames from the videos, and selecting suitable frames where the garment is clearly visible.

2.Developing the garment segmentation dataset and model

Creating polygon annotations for the selected frames using the Roboflow platform, incorporating background images (without garments), and training a segmentation model to identify the garment area.

3.Evaluating runtime frame suitability and measurement stability

Identifying frames unsuitable for measurement using detection-zone and garment-geometry checks, and verifying the stability of dimensions across consecutive frames before confirming a measurement.

4.Obtaining physical measurements based on garment type

Capturing length and width in pixels based on specific measurement points for T-shirts, shirts, and trousers, and converting these values into centimeters using a calibration factor suited to the camera setup.

5.Size estimation and result presentation

Estimating the size by comparing the obtained dimensions against the size chart relevant to the selected garment type, and displaying an annotated image-featuring measurements, size details, and measurement lines-on the user interface.

6.Managing measurement records and garment counting

Saving confirmed garment measurements and size-wise counts, employing removal and re-arming logic to prevent the same garment from being counted repeatedly, and maintaining measurement history and production-order records.

7. Evaluating the developed component

Testing the performance of the segmentation model, calculating measurement error by comparing system measurements against reference measurements, and evaluating the impact of frame-stability checks as well as processing time.

1.6 Research Questions or Hypotheses

This study is guided by research questions. The primary research question is: How accurately and consistently can garment length and width be measured by combining frame-suitability checks, garment segmentation, and calibration under a controlled overhead camera setup?

The following questions address specific aspects of this investigation:

1. How accurately does the segmentation model trained on frames collected from garment manufacturing environments identify the garment regions and boundaries required for dimensional measurement?
2. To what extent do detection-zone and garment-geometry checks reduce the acceptance of incomplete or unsuitable frames for measurement?
3. Compared with single-frame measurements, does checking measurement stability across three consecutive frames reduce measurement variability and error relative to reference measurements?
4. How much do the measured garment dimensions differ from reference measurements when using a fixed camera height of 100 cm and a defined calibration? How does this error vary across T-shirts, shirts, and trousers?
5. To what extent do size estimates obtained from system-generated dimensions agree with those obtained by applying reference measurements to the same garment-specific size chart?
6. How effectively does the garment-removal and re-arming logic prevent duplicate counting while allowing subsequent garments to be measured and counted?
7. What is the processing time between receiving an image at the backend and returning a measurement response, and is this time compatible with the intended scanning interval?

1.7 Scope Limitations and Assumptions

Scope

This research component focuses on measuring the length and width of T-shirts, shirts, and trousers from camera images and estimating garment size from these dimensions. Garments are placed flat on a measurement surface and captured using an overhead camera positioned at a fixed height of 100 cm.

The component includes garment segmentation, detection-zone and geometry checks, measurement-stability verification across three consecutive frames, pixel-to-centimeter conversion, and size estimation using garment-specific size charts. It also

provides measurement overlays, size-wise counts, garment-removal and re-arming logic, measurement history, and production-order records. The garment type is selected through the frontend and supplied to the backend; automatic garment-type classification is outside the scope of this component.

The measurement task concerns flat garment length and width. Human body measurement, three-dimensional garment reconstruction, fabric-defect detection, and verification of fit on an individual wearer are outside the research scope.

Limitations

The measurement process depends on calibration for a defined camera setup. Changes in camera height, viewing angle, zoom, or image-processing scale may affect measurement accuracy. The current configuration uses a calibration of 13.6 pixels per centimeter at an image resolution of 1920 × 1080. Changes to capture conditions require verification of the calibration and pixel-coordinate transformations.

Garment folds, stretching, occlusion by hands or other objects, incomplete visibility, and variations in lighting may affect segmentation and measurement. Model performance may also vary for garment designs, materials, and backgrounds that are insufficiently represented in the training dataset.

Size estimation is restricted to the configured size charts. The current charts contain reference entries from S to XXL for T-shirts and trousers, and from XS to 3XL for shirts. As size definitions may differ between brands and product specifications, the estimated label is not a universal manufacturer size or a guarantee of fit for an individual wearer.

The requirement for stability across three consecutive frames also means that performance on garments moving rapidly through the camera view requires separate evaluation. Processing time may vary according to the available hardware and workload.

Assumption

The camera is assumed to remain perpendicular to the measurement surface at a fixed height and zoom setting. Each garment is assumed to be placed flat without substantial

stretching or folding, with only one garment present within the detection zone at a time.

The system assumes adequate lighting and visibility of the garment boundaries, and that the operator selects the correct garment type. It also assumes that the calibration corresponds to the capture setup and that the configured size chart is an appropriate reference for the garments being evaluated.

For evaluation, reference measurements are assumed to follow the same measurement definitions and endpoints used by the system.

1.8 Individual Contribution

My individual contribution to this group research project is the development of the garment dimension measurement and size estimation component. This work focuses on selecting suitable garment frames, developing a garment segmentation model, and incorporating the trained model into a calibrated measurement workflow.

For data collection, I visited garment manufacturing environments and captured video footage using an overhead camera setup positioned at a fixed height of 100 cm. I extracted frames from the collected videos and selected images with clearly visible garment boundaries that were suitable for segmentation annotation.

For dataset preparation, I used Roboflow to annotate garment boundaries with polygons. I also included background images without garment annotations and prepared multiple segmentation dataset versions for model training. The resulting trained model was incorporated into the garment measurement component.

The measurement-related contribution includes garment-type-specific measurement logic, pixel-to-centimeter calibration, and size estimation using configured size charts. The runtime applies detection-zone and geometry checks to reduce the acceptance of unsuitable frames and verifies measurement stability across three consecutive frames before confirming a measurement. Garment-removal and re-arming logic is used to prevent repeated counting of the same garment.

The component connects its frontend and backend functionality to present dimensions, estimated sizes, and measurement overlays to the operator. Measurement history, size-wise counts, and production-order records are included as supporting functionality.

My contribution within the complete project is limited to the garment measurement and size estimation component. Production Monitoring, Rework Counter, and Garment Style and Color Detection are separate team components, and their development is not claimed as part of my individual contribution.

1.9 Thesis Structure

This thesis begins by introducing the research context and problem. It reviews relevant literature and identifies the research gap before describing the methods used for data collection, dataset preparation, model training, and evaluation. It also covers component design, implementation, integration, testing, and commercialization considerations. The evaluation results are then presented and discussed in relation to the research objectives, contributions, and limitations. The concluding section summarizes the findings and identifies future development directions. The references and appendices provide the cited sources and supplementary research evidence.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

This literature review is organized as a focused narrative review of computer vision methods relevant to garment dimension measurement and size estimation. Its purpose is to compare previous research that informs the design decisions and evaluation approach of the proposed individual component.

The scope includes garment segmentation, the identification of frames suitable for measurement, garment boundary and measurement point extraction, the conversion of pixel measurements into physical dimensions, and size estimation using predefined size charts. Measurement stability across consecutive frames and the rearming of the measurement process following garment removal are also considered where relevant.

The principal criteria for selecting studies are their relevance to image-based garment measurement, the clarity of their methodological descriptions, and the availability of evaluation evidence. Peer-reviewed research papers serve as the main sources of research evidence, while official technical documentation supports explanations of implementation-specific functionality. Studies on human body measurement, virtual fitting, and fabric defect detection are included only where their methods directly inform the proposed component.

Selected studies are compared according to garment categories, dataset characteristics, camera arrangements, segmentation methods, calibration requirements, measurement error, and processing performance. Reported results are interpreted in the context of each study's data and experimental conditions rather than treated as directly equivalent. This comparison supports the identification of design requirements and the research gap relevant to garment measurement under a fixed overhead camera arrangement.

2.2 Domain and Theoretical Foundations

Garment measurement and segmentation

Image-based garment measurement determines distances between defined locations on a garment placed on a flat surface. These locations differ between T-shirts, shirts,

and trousers; consequently, bounding-box dimensions alone may not represent the required garment measurements.

Instance segmentation provides a pixel-level mask for each detected object. The proposed component uses YOLOv8 segmentation to separate the garment from its background and support geometric measurement. However, successful segmentation does not independently establish measurement suitability. Folding, occlusion, and incomplete visibility must also be considered before dimensions are accepted. Instance masks are distinct from bounding-box detections, as illustrated by established instance-segmentation research [4] and the Ultralytics segmentation interface [5].

Spatial calibration

Image distances are expressed in pixels and require calibration for conversion into physical units. Camera geometry and lens distortion influence this relationship. The component uses a fixed overhead camera arrangement at a height of 100 cm and applies a calibrated pixels-per-centimetre scale: Camera calibration and distortion are described in OpenCV documentation [8] and Zhang's calibration work [9]; these references support the geometric background, not a claim that Zhang's full calibration procedure was implemented here.

$$d_cm = d_px / s_px_per_cm$$

Dimension_cm denotes the physical distance, Dimension_px denotes the corresponding image distance, and pixels_per_cm denotes the calibrated scale. The stored reference scale is 13.6 pixels/cm at 1920 × 1080. The backend adjusts the scale for the submitted frame resolution. This assumes consistent camera geometry and a compatible image scale; arbitrary crops, zoom changes or changes in the measurement plane require calibration checks. Marker-free operation therefore still requires initial calibration.

Temporal stability and size estimation

Consistency across consecutive frames is used to reduce temporary measurement variation. Stability concerns repeatability and must be distinguished from accuracy, which requires comparison with reference measurements.

Accepted dimensions are mapped to a predefined size chart for the selected garment type. The output is therefore a chart-based size estimate. A confirmed measurement remains locked until garment removal is established, after which the component is rearmed for the next garment. This behavior controls repeated counting across consecutive frames.

2.3 Related Research

Serrat, Lumbreras and Ruiz used garment-contour interpretation and learned regressors for preshipment measurement [3]. Their study reported 1.17 cm mean absolute error (MAE) for jackets and 1.22 cm for pants using 130 jacket images and 98 pants images. These results provide context for dimensional-error reporting; they are not a matched baseline for the present prototype.

Paulauskaite-Taraseviciene et al. investigated automatic garment segmentation and measurement-point extraction [1]. Their separation of garment localisation from dimensional measurement informs the present combination of segmentation masks and geometric processing.

Kim et al. combined image-based measurement-point estimation with LiDAR point-cloud distances [2]. They reported average relative errors of 1.59% and 2.08% in two environments. The present component uses a fixed overhead arrangement and calibrated pixel scale, so hardware, metrics and acquisition conditions differ and do not support a direct performance ranking.

These studies provide complementary foundations for contour interpretation, mask-based measurement, and spatial calibration. Their results must be interpreted within their respective datasets and experimental conditions. They do not directly establish the performance of the proposed live measurement workflow. Table 2.1 summarizes their relevance to this component.

Table 2.1 summarises comparison of garment measurement approaches.

Table 2.1: Comparison of garment measurement approaches

Source	Approach	Evidence or scope	Relevance and comparison limit
Paulauskaite-Taraseviciene et al. [1]	Segmentation and measurement-point extraction	Automated garment measurement study	Supports separate localisation and geometry stages; different data and workflow.
Kim et al. [2]	Image keypoints with LiDAR distances	Relative error 1.59% and 2.08% in two environments	Depth-based alternative; errors are not directly comparable with this study's MAE.
Serrat et al. [3]	Contour interpretation and learned regression	130 jacket and 98 pants images; MAE 1.17 and 1.22 cm	Supports dimensional-error evaluation; different garment categories.
Present component	YOLO masks, fixed scale and temporal confirmation	Five tape-reference records; preliminary MAE	Controlled setup; no matched baseline or production-scale validation.

Mask R-CNN provides a research foundation for instance-level mask prediction [4]. It is cited as related segmentation work; the implemented model is YOLOv8n-seg and no Mask R-CNN benchmark was conducted.

2.4 Existing Systems Products and Techniques

Existing measurement approaches differ according to their measurement target and operating conditions. Manual garment measurement uses defined measurement locations and a measuring tape. It provides a reference approach for evaluating the proposed component, provided that manual and automated measurements follow the same endpoint definitions and garment preparation procedure.

Automated research systems use garment contours, segmentation masks, measurement points, or depth information. Their compatibility with the proposed workflow depends on camera requirements, calibration assumptions, garment categories, and output definitions. Reported accuracy alone is therefore insufficient to establish equivalence between systems.

Size Stream describes commercial body-scanning and measurement technology [14]. Its target is the human body, whereas this component measures flat garments. Vendor information is treated as product context rather than independent evidence of garment-measurement accuracy.

The proposed component focuses on controlled flat-garment measurement. Its outputs comprise calibrated dimensions, a size estimate based on a garment-specific chart, and a confirmed garment count.

2.5 Gap Synthesis and Conceptual Framework

Research gap

Previous research has established contour-based, segmentation-based, and depth-assisted approaches to automated garment measurement. The gap addressed by this research concerns their application within the proposed operational context.

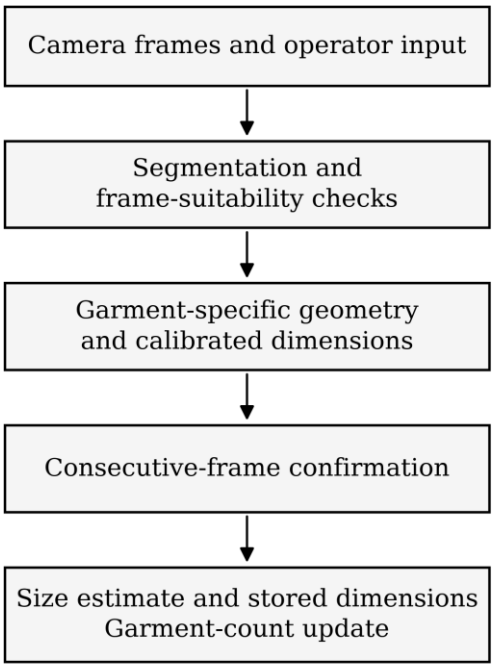
The component combines suitable-frame acceptance, calibrated dimension extraction, size-chart mapping, and repeated-count prevention under a fixed overhead camera arrangement at a height of 100 cm. The reviewed studies do not establish the performance of this combined workflow for the selected garment categories and project-specific acquisition conditions. Independent evaluation is therefore required. The intended contribution is the integration and evaluation of these methods within a defined workflow.

Conceptual framework

The framework receives camera frames and an operator-selected garment type, together with calibration settings and a relevant size chart. Segmentation produces a garment mask, which is assessed for geometric suitability and consistency across consecutive frames. Figure 2.1 illustrates this workflow.

For an accepted presentation, garment-specific pixel dimensions are converted into centimeters using the calibrated scale. These dimensions are mapped to the relevant size chart. The confirmed measurement is recorded and the garment count is updated. The process remains locked until garment removal is established and is then rearmed.

Evaluation considers measurement accuracy, temporal stability, size-estimation agreement, repeated-count prevention, and processing time. These outcomes are assessed separately because stable measurements do not necessarily represent accurate physical dimensions.



Controlled setup: overhead camera at 100 cm; initial calibration required.

Figure 2.1: Conceptual framework of the garment-measurement workflow

3 RESEARCH METHODOLOGY

3.1 Research Design

This research adopts an applied, artefact-centered design to develop and evaluate a computer vision component for garment dimension measurement and size estimation. The component combines garment segmentation, calibrated geometric measurement, temporal confirmation, size-chart mapping, and repeated-count prevention.

The methodology comprises two stages. The development stage involves collecting garment footage, preparing annotated datasets, and training a segmentation model. The evaluation stage examines the component under a controlled overhead camera arrangement, using reference measurements and defined garment-presentation scenarios.

Model-level segmentation performance and system-level measurement performance are evaluated separately. This distinction allows errors in garment localization to be considered independently from errors arising during geometric measurement, calibration, or size estimation.

3.2 Data Participants or Study Materials

The primary study materials comprise videos collected in garment-manufacturing environments, extracted video frames, selected garment images, and background-only images. Footage was captured using a camera positioned at a fixed height of 100 cm above the measurement surface.

Selected images were annotated in Roboflow using polygons representing individual garment boundaries. Background-only images were included without garment annotations rather than assigned a separate garment class. Multiple dataset versions were prepared during model development. Roboflow dataset versions preserve preparation settings and exported data for subsequent training [10].

The documented physical comparison consists of five T-shirt and shirt records measured with a tape and the component. Records 1 and 2 are repeated trials of the same garment. The identity relationship among the other records is not fully documented, so a count of five independent garments is not asserted. Trousers are

supported by the implementation but are not represented in this five-record comparison. No structured operator study is included.

The final dataset configuration specifies one class, new-garment, and Roboflow version 5. The image counts supplied by the researcher are shown in Table 3.1. Counts of source videos, distinct garment instances and background-only images were not supplied separately. Split independence by source video or garment has not been verified.

Table 3.1 summarises segmentation dataset partitions.

Table 3.1: Segmentation dataset partitions

Partition	Images	Share	Configured path
Training	1070	82.63%	train/images
Validation	141	10.89%	valid/images
Test	84	6.49%	test/images
Total	1295	100%	Single class: new-garment

3.3 Experimental or Study Procedure

Data preparation began with establishing the fixed-height camera arrangement and collecting garment videos. Frames were extracted and suitable images were selected for polygon annotation. Background-only images were added, and dataset versions were prepared for segmentation training. Figure 3.1 documents the overhead arrangement.

The supplied training configuration specifies YOLOv8n-seg, 50 epochs, batch size 2, image size 640, seed 42 and validation enabled. The training partition supports learning and the validation partition supplies the logged metrics [6], [7]. Although 84 test images are listed, no independent test-set result was supplied. The researcher confirmed that the deployed best_model.pt originated from this training run; the precise saved-checkpoint epoch was not independently established.

Table 3.2 summarises recorded model training configuration.

Table 3.2: Recorded model training configuration

Setting	Recorded value
Task	Instance segmentation
Initial model	yolov8n-seg.pt
Epochs configured and logged	50
Batch size	2
Training image size	640
Seed	42
Optimizer setting	auto; resolved optimizer not recorded in supplied evidence
Device argument	0; graphics processing unit (GPU) model not established by args.yaml
Validation split	val
Dataset directory	garment_seg_100cm_clean_v5
Evidence	test01/args.yaml and test01/results.csv

For the preliminary dimensional comparison, manual values were obtained using a tape, as confirmed by the researcher. Each recorded manual width and length was paired with the corresponding system output in centimetres. The handwritten sheet was transcribed without swapping its width and length columns. Exact garment landmarks, camera conditions for each record and manual repeat-measurement uncertainty were not separately documented, and this limits the interpretation of the comparison.

A broader evaluation protocol should include suitable and unsuitable presentations, prolonged garment presence, and removal followed by a new presentation. Folding, occlusion, movement and incomplete visibility are relevant conditions. These scenarios are proposed extensions and are not reported as completed experiments.

A matched comparison of single-frame measurement and consecutive-frame confirmation remains future work. The supplied records do not form such a comparison and do not establish the effect of temporal confirmation on accuracy or variation.

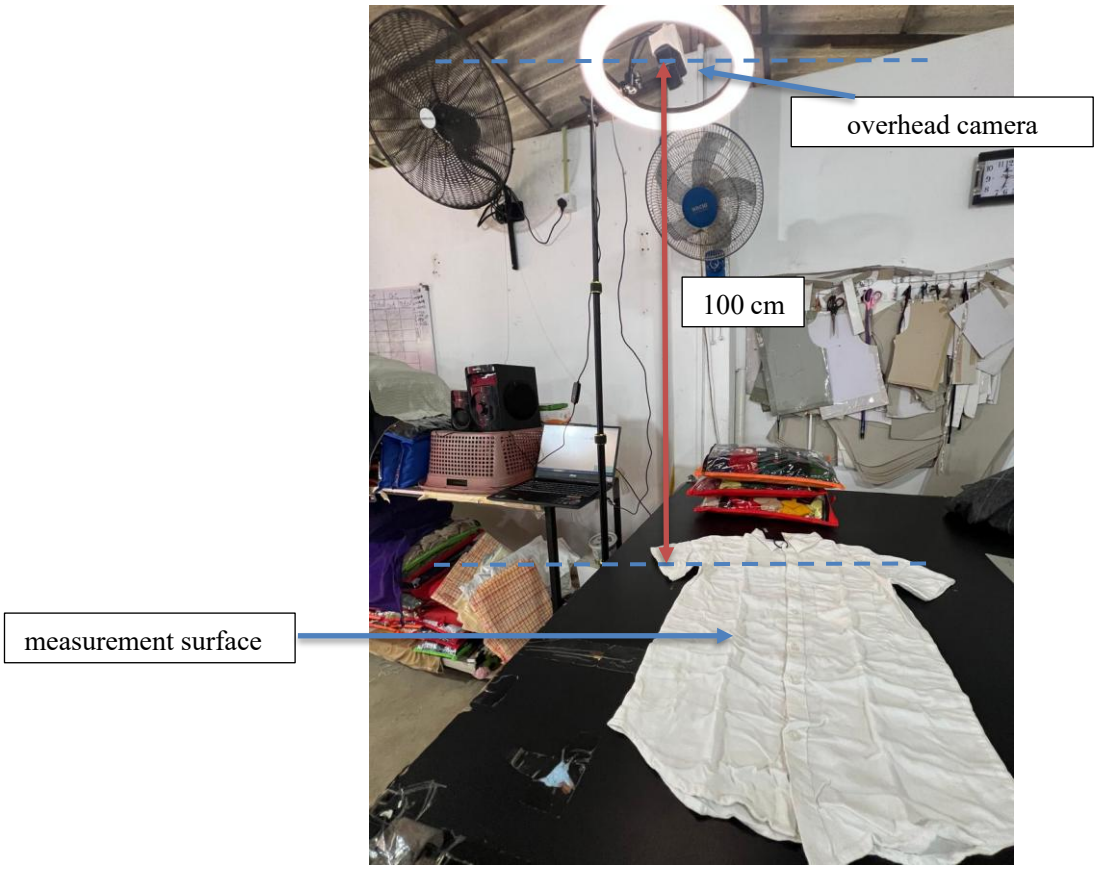


Figure 3.1: Fixed-height overhead camera arrangement used for garment data acquisition and measurement (Original in Color)

3.4 Measures and Evaluation Metrics

Evaluation addresses segmentation quality, measurement accuracy, temporal stability, size-estimation agreement, counting behavior, and processing time. These outcomes are assessed independently because performance in one does not establish performance in another.

For each dimension, absolute error is |system - manual|. Mean absolute error (MAE) is calculated as MAE = sum(|system − manual|)/n. Root mean square error (RMSE) is calculated as RMSE = sqrt(sum((system − manual)^2)/n). Both are reported in centimetres across the five recorded comparisons. This record-level summary includes two observations of one garment and must not be interpreted as a summary of five independent garments.

Precision, recall, mean average precision (mAP) at 0.50 and mAP@0.50–0.95 are taken from the validation log and reported separately for boxes and masks [7]. These

model metrics do not measure centimetre accuracy. Reference size labels, count-event ground truth and timing records were not supplied, so size agreement, counting reliability and latency are not calculated.

Table 3.3 summarises evaluation measures and available evidence.

Table 3.3: Evaluation measures and available evidence

Measure	Definition or purpose	Available evidence
Mask and box metrics	Precision, recall and mAP from validation	50-epoch comma-separated values (CSV) log; no independent test result
Absolute error	|system dimension − tape dimension|	Five paired records
MAE and RMSE	Record-level dimensional summaries in cm	Calculated separately for width and length
Repeatability	Variation under repeated presentations	Only two identified trials of one garment; broader evaluation needed
Size agreement	System label compared with independent reference	Reference labels absent
Counting and latency	Ground-truth count events and defined timing	Dedicated records absent

3.5 Data Analysis

The training log was inspected for final-epoch and peak validation metrics. The five tape-reference records were transcribed and absolute errors, MAE and RMSE were calculated separately for width and length. Estimated size labels are retained as outputs rather than treated as reference classifications.

No statistical significance test, matched temporal baseline, latency distribution or count-error rate is reported because the necessary observations were not supplied. Error patterns are discussed as preliminary findings rather than evidence of generalisation.

Repeated records from the same garment are related observations. The pooled five-record errors describe this recorded comparison only; garment-level uncertainty and independent-sample performance require a larger, identified evaluation set.

3.6 Ethics Privacy Security and Responsible AI

Data acquisition and use must follow the permissions applicable to each manufacturing environment. Only information necessary for garment measurement should be retained. Identifiable people, employee details, order information, and confidential factory material should be excluded or anonymized where applicable.

Access to footage, annotations, and model artefacts should be restricted. Private data and credentials must not be included in public repositories. Uploading images to third-party annotation platforms requires consideration of the applicable permissions and dataset visibility settings. Permission and approval records should be reported only where they have actually been obtained.

Responsible operation requires unsuitable or unsupported presentations to be rejected rather than assigned unjustified measurements. The component's outputs remain subject to its calibration conditions, dataset coverage, and size-chart definitions. These limitations, together with relevant AI-assisted development and writing activities, should be disclosed accurately.

3.7 Reproducibility and Research Artefacts

Reproducibility requires records of dataset versions, annotation formats, data partitions, training configurations, model checkpoints, and software dependencies. The selected checkpoint should be linked to its training dataset and configuration.

The camera arrangement should be documented through height, orientation, resolution, zoom settings, and calibration scale. Evaluation artefacts should include reference measurements, presentation scenarios, analysis scripts, and recorded outputs. These records support the interpretation and repetition of the evaluation.

Source code, size charts and configuration records complete the reproducibility artefacts; restricted footage requires permitted anonymised evidence.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The component is designed to measure flat T-shirts, shirts, and trousers and estimate their sizes using garment-specific charts. The operator selects the garment type, while the component evaluates the submitted image, extracts dimensions, and confirms measurements across consecutive frames.

Functional requirements cover measurement, size estimation, counting, and record management. Quality requirements concern repeatability, clear feedback, processing performance, and maintainability. Operation assumes a calibrated overhead camera arrangement at a fixed height of 100 cm. Numerical acceptance thresholds for accuracy and processing time must be established through the evaluation protocol.

Table 4.1 summarises requirements and evidence boundaries.

Table 4.1: Requirements and evidence boundaries

Requirement	Implemented behaviour	Validation status
Dimension measurement	Mask-based geometry and calibrated conversion	Five-record preliminary comparison
Garment type input	Operator selects supported category	Not predicted as three segmentation classes
Size estimation	Garment-specific chart mapping	Reference size agreement not measured
Temporal confirmation	Three stable observations	Effectiveness not benchmarked
Repeated-count control	Locked result until removal and rearming	Controlled positive count trials needed
Record management	SQLite measurement and order history	Interfaces and existing software checks
Performance and usability	Local desktop workflow	Latency and operator study outstanding

4.2 Component Architecture within the Complete Solution

The complete solution uses a shared React interface within an Electron desktop application. The Electron main process manages backend startup and runtime

configuration, while the preload layer provides controlled communication between the interface and desktop functionality.

The individual garment-measurement component operates through a separate FastAPI service. It loads the segmentation model, processes measurement requests, applies calibration and size-chart configuration, and manages measurement records through SQLite. Figure 4.1 shows its place in the shared application.

The production-monitoring, rework, and style/color components use separate backend services within the same desktop application. The current architecture integrates these features through the shared application shell rather than direct cross-component backend calls. In particular, the measurement component receives an operator-selected garment type rather than an automatically supplied classification from the style/color component.

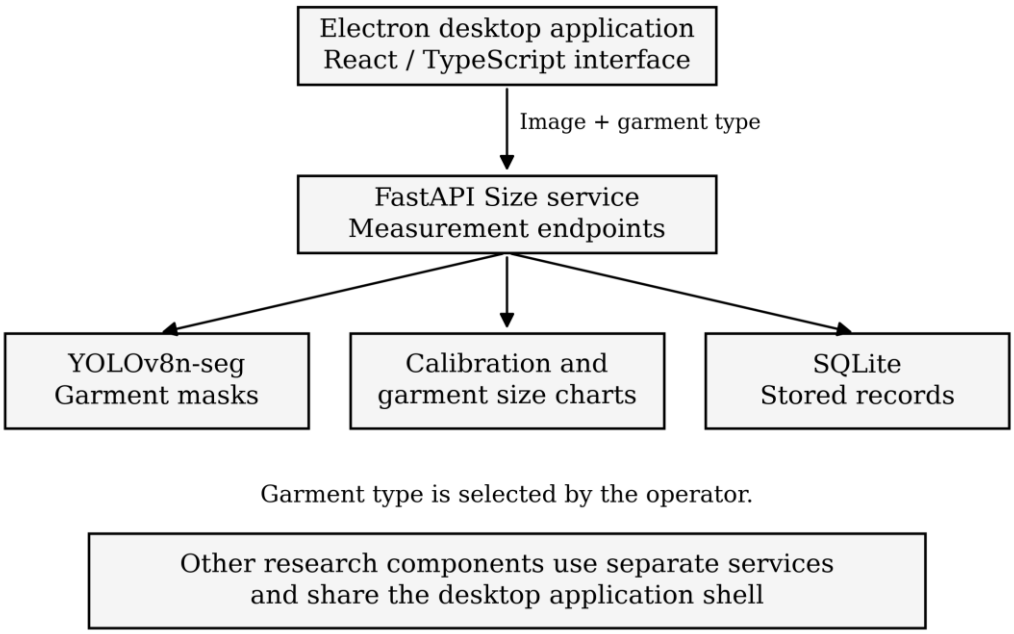


Figure 4.1: Architecture of the garment-measurement component

4.3 Design Alternatives and Decisions

The design considers alternatives for garment localisation, physical-distance estimation, measurement confirmation, and data storage. The selected approaches

reflect the controlled camera arrangement and the requirements of the individual component.

Segmentation-based geometry supports garment-specific measurement locations more directly than bounding-box dimensions alone. A calibrated fixed-height arrangement avoids reliance on depth-sensing hardware or a marker in each operational frame, while imposing consistent camera and measurement-plane requirements.

Consecutive-frame confirmation is selected to control temporary measurement variation. Its effect on stability and confirmation time requires evaluation. Table 4.2 records the principal choices and their trade-offs.

Table 4.2: Design choices and trade offs

Decision	Reason	Constraint
Segmentation masks	Support geometry beyond the enclosing box [5]	Mask errors can affect measurement locations
Fixed-height scale	Convert pixel distances without a runtime marker	Camera geometry and measurement plane must remain compatible
Operator garment type	Select the relevant geometry and chart	Depends on correct selection
Three-frame confirmation	Require consecutive stable candidates	Benefit and added delay need controlled evaluation
SQLite storage	Local embedded persistence [12]	Not evidence of multi-user production deployment

4.4 Detailed Design

The interface submits an image file and a selected garment type to the /measure endpoint. The backend performs segmentation and derives the geometry required for garment-specific measurement. Frames without a detected garment or an acceptable presentation do not produce a confirmed measurement.

For acceptable candidates, pixel distances are calculated between defined measurement locations and converted into centimetres using the calibration scale. Consecutive-frame checks determine whether the measurement can be confirmed. The resulting dimensions are then interpreted using the size chart for the selected garment type. Figure 4.2 summarises this processing pipeline.

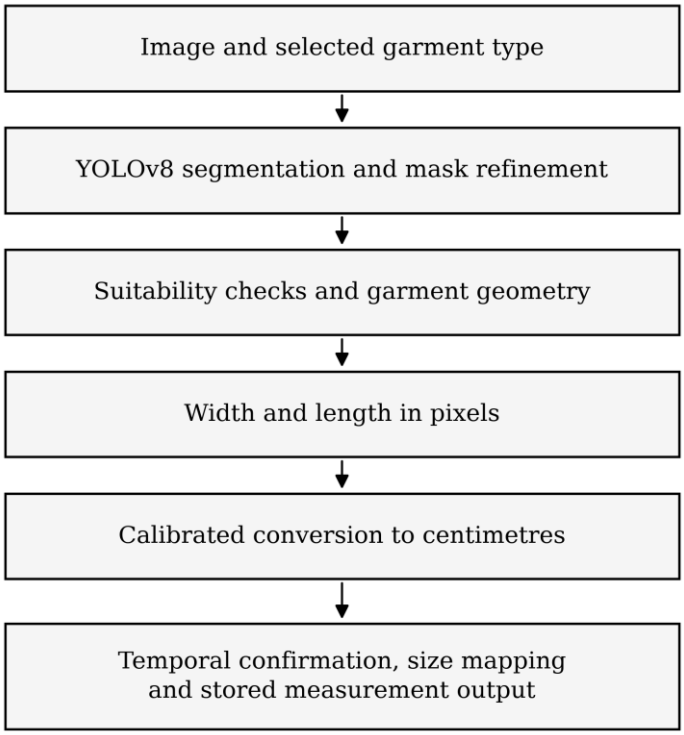


Figure 4.2: Garment segmentation and dimension-measurement pipeline

Source: schematic of the implemented workflow described in Section 4.4.

Measurement state and counting

The component follows a state-based process: awaiting a garment, assessing a candidate, and retaining a confirmed measurement. Candidate observations contribute to confirmation, while a confirmed result remains locked during continued garment presence. Confirmed removal returns the process to its initial state.

This design separates frame processing from garment-count events. Multiple images of the same garment therefore do not automatically generate multiple counts. Figure 4.3 shows the confirmation and rearming states; the diagram describes implemented logic rather than a measured reliability result.

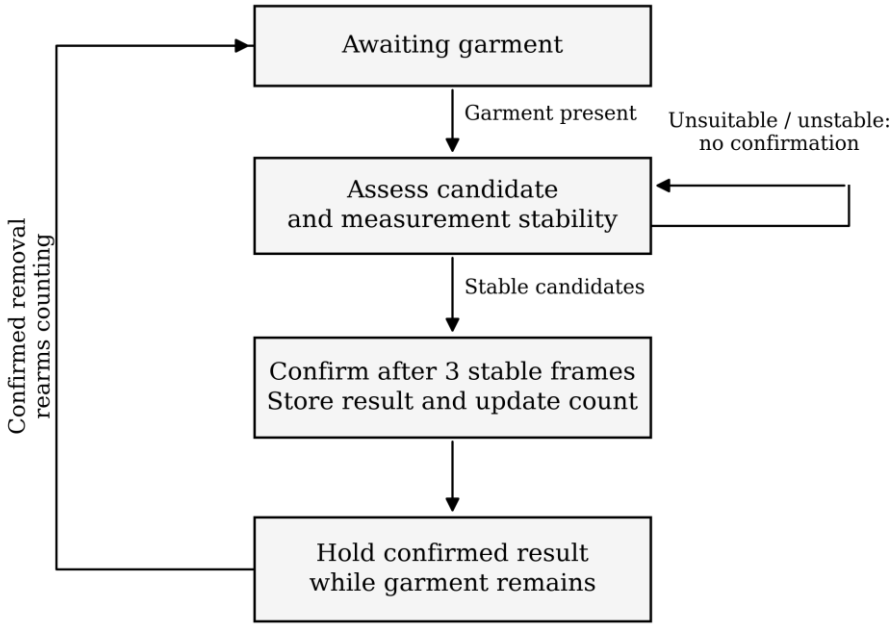


Figure 4.3: Measurement confirmation and garment-removal logic

Source: component state logic; three-frame confirmation from the runtime configuration.

Configuration and persistence

Calibration settings and size charts are maintained separately from model weights. SQLite supports persistent measurement, order, and count-related records. The application programming interface (API) also provides health information and endpoints for retrieving and managing component records.

The interface communicates the current measurement state and displays available dimensions and size estimates. Processing logic and persistence remain in the backend, allowing the interface to focus on capture, operator input, and feedback

4.5 Standards and Professional Practice

The component uses Hypertext Transfer Protocol (HTTP) endpoints, JavaScript Object Notation (JSON) responses, and multipart image uploads. FastAPI provides OpenAPI documentation to describe available operations and request formats,

supporting interface consistency and integration. These interface capabilities are documented by FastAPI [11].

Professional practice includes version control, dependency documentation, configuration management, and traceable dataset and model versions. Measurement units, endpoint definitions, operating assumptions, and failure states should remain explicit throughout implementation and evaluation.

Private manufacturing data and credentials must be excluded from public repositories. Software, dataset, and model licenses should be reviewed, with appropriate attribution retained. Design and testing evidence should support any claims concerning reliability, security, or performance. The component is not presented as formally certified against an external standard unless such certification has been obtained.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The application is implemented as an Electron desktop platform with a React and TypeScript interface. The garment-measurement backend uses Python, FastAPI, and Uvicorn to provide local application services. FastAPI documents request handling and generated API documentation [11], while Electron separates main and renderer responsibilities [13].

YOLOv8 segmentation supplies garment masks, while OpenCV and NumPy support image processing and geometric measurement. SQLite stores measurement and order-related records. Calibration settings and garment-specific size charts are maintained in JSON configuration files. The technologies are supported by the segmentation, OpenCV and SQLite documentation [5], [8], [12].

The supplied frontend package requires Node.js version 22.12 or later and uses npm start to launch the desktop application. A shared Python environment supports the backend services. Software dependency records and configuration files provide the basis for reproducing the development environment.

The supplied health response shows central processing unit (CPU) execution, inference image size 640, camera height 100 cm, reference scale 13.6 pixels/cm at 1920 × 1080, three required stable frames and SQLite persistence. These are runtime settings and are distinguished from the training device argument in Table 3.2.

5.2 Component Implementation

The interface captures camera frames and submits them with the operator-selected garment type to the measurement backend. In desktop operation, requests pass through the Electron communication layer to the local Size API.

The backend loads the trained YOLOv8 segmentation checkpoint and processes incoming images to obtain garment masks. Mask refinement, orientation handling, suitability checks, and garment-specific geometry support dimension extraction. Pixel distances are converted into centimetres using the calibrated scale, and accepted dimensions are interpreted through the relevant size chart. The researcher confirmed

that models/garment-size/best_model.pt is the checkpoint deployed from the supplied test01 training run. Its exact validation score should be established by validating that saved checkpoint rather than assigning the final-epoch or peak score automatically.

The current configuration requires three consecutive observations for measurement confirmation. A confirmed result remains locked until garment removal is established, controlling repeated counting. SQLite supports measurement and order-related persistence, while the interface presents measurement feedback, counts, and historical records.



Figure 5.1: Confirmed garment dimensions and estimated size in the application

Figure 5.1 shows the measurement overlay and recorded outputs in the implemented interface. The screenshot illustrates the application; the tape-reference comparisons in Chapter 6 are a separate source of dimensional evidence.

5.3 Integration with the Other Components

The four research components are presented through a shared React and Electron application. The Electron main process manages feature-service startup, runtime configuration, and local request routing.

The Size component retains its own backend, trained model, and database. Integration occurs through the shared desktop platform rather than direct backend dependencies between the research components. The selected garment type is supplied by the operator; it is not automatically received from the style/color component.

Frontend compilation checks successfully built the main process, preload layer, and renderer. These checks support the consistency of the application structure but do not establish complete live-camera or hardware integration.

5.4 Verification and Software Testing

Earlier verification recorded frontend type checking, 57 automated tests and compilation of the Electron main, preload and renderer targets. These outcomes are retained as previously reported software checks; fresh execution logs were not supplied with this revision. They are not treated as new independently reproduced test results.

The newly supplied health response confirms healthy status and successful model loading on CPU in the Size backend. It exposes calibration, model-path, image-size and persistence settings, as shown in Figure A.2.

Earlier checks reported successful responses for count, measurement-history and order endpoints, a blank-image negative control without detection or count increment, and expected errors for invalid inputs. Those checks are distinguished from the newly supplied health screenshot and the five physical measurement records.

The evidence supports selected software behaviours and a preliminary physical dimension comparison. It does not establish complete API test coverage, removal detection reliability or duplicate-count prevention under repeated real-garment presentations. Table 5.1 distinguishes the sources and limits of verification.

Table 5.1 summarises software verification and runtime evidence.

Table 5.1: Software verification and runtime evidence

Check	Evidence status	Interpretation
Model loading and health	New screenshot: healthy; model_loaded true	Confirms current service and model loading
CPU and calibration settings	New /health response on port 8013	Documents runtime configuration
Frontend tests and compilation	Previously reported: type check and 57 tests passed	Fresh logs not supplied for this revision
API response and invalid-input checks	Previously reported software checks	Not newly rerun in this revision
Dimensional comparison	Five tape and system records	Physical errors summarised in Chapter 6
Counting, latency and usability	No dedicated evaluation records supplied	Cannot assign numerical performance results

5.5 Deployment and Operation

The application was launched in the Windows development environment using npm start. The supplied startup evidence shows compilation of the Electron main process, startup of the renderer development server, and launch of the Electron application. The measurement interface displays a connected-backend indicator.

Following startup, the operator selects the Garment Size Detection workstation, creates or resumes an order, and configures the camera and garment type. Measurement operation requires the fixed-height camera arrangement and calibration conditions to be maintained. Each garment is placed flat and removed after measurement confirmation before the next garment is presented.

The available evidence demonstrates development-mode startup and the displayed backend connection state. It does not establish packaged-installer deployment or measurement accuracy.

6 RESULTS AND VALIDATION

6.1 Research Results

The supplied test01 training artefacts include args.yaml, results.csv and diagnostic images. The CSV contains 50 epochs of box and mask validation metrics. Table 6.1 reports the final logged epoch; Table 6.2 reports the peak mAP@0.50–0.95 epoch separately. These are validation results, not independent test results or dimensional accuracy percentages.

A preliminary physical comparison uses five paired tape and system measurement records for T-shirts and shirts. Records 1 and 2 are repeated trials of the same garment. The researcher confirmed all values in centimetres and retained the recorded width/length order for shirts. Table 6.3 contains the raw comparison and Table 6.4 contains the derived errors.

The first two system widths differ by 0.50 cm and lengths by 0.95 cm for the same garment. Two trials are insufficient to establish repeatability or compare temporal confirmation against a single-frame baseline.

All ten recorded absolute dimensional errors are below 1.5 cm. No prior tolerance threshold was documented, so this observation is not described as passing a predefined acceptance criterion. Estimated sizes S, S, M, S and M are outputs; independent reference sizes were not supplied.

Figure 6.1 plots segmentation training and validation loss and mask validation mAP across the 50 logged epochs. These curves describe the supplied training run, not independent test-set results or the exact validation score of the deployed checkpoint.

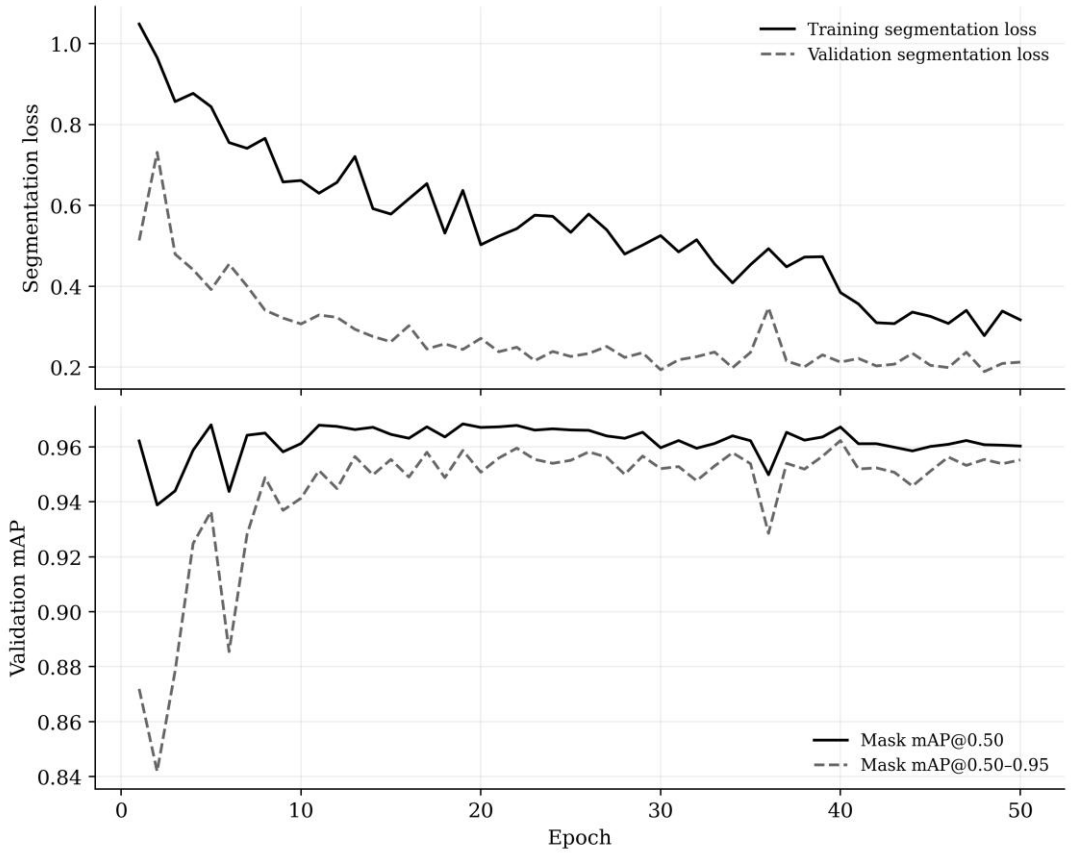


Figure 6.1: Segmentation training loss and validation performance across epochs

Source: test01/results.csv; values plotted directly from the supplied 50-epoch log.

Table 6.1 presents final epoch validation results.

Table 6.1: Final epoch validation results

Metric	**Bounding box**	**Segmentation mask**
Precision	0.97707	0.97707
Recall	0.97037	0.97037
mAP@0.50	0.96033	0.96033
mAP@0.50–0.95	0.95222	0.95529

Table 6.2 summarises peak validation mAP and reporting boundaries.

Table 6.2: Peak validation mAP and reporting boundaries

Result	Epoch	Value	Interpretation
Box mAP@0.50–0.95	40	0.96142	Peak in supplied CSV
Mask mAP@0.50–0.95	40	0.96232	Peak in supplied CSV
Deployed checkpoint	Not known	Not assigned	Same run confirmed by researcher; exact checkpoint validation not supplied
Independent test evaluation	Not supplied	Not assigned	84 test images listed; evaluation result absent

Table 6.3: Tape reference and system dimensions in centimetres

ID	Type	Tape width	Tape length	System width	System length	Size
1	T-shirt	45.72	62.04	45.50	61.10	S
2	T-shirt	45.72	62.04	46.00	62.05	S
3	T-shirt	49.53	68.50	49.50	67.03	M
4	Shirt	68.58	45.72	69.91	45.03	S
5	Shirt	71.12	48.26	72.11	48.00	M

Records 1 and 2 concern the same garment. Shirt width and length values follow the original sheet and the researcher's explicit confirmation. The larger recorded width than length makes landmark documentation important; exact measurement landmarks were not supplied. These records are not presented as five independent garments.

Table 6.4: Dimensional errors across the five recorded comparisons

Record or metric	Width error cm	Length error cm
1	0.22	0.94
2	0.28	0.01
3	0.03	1.47
4	1.33	0.69
5	0.99	0.26
MAE	0.570	0.674
RMSE	0.759	0.847
Maximum absolute error	1.33	1.47

Figure 6.2 compares absolute width and length errors for the five recorded tape-reference comparisons. Records 1 and 2 are repeated trials of one garment; the plot does not represent five independent garments.

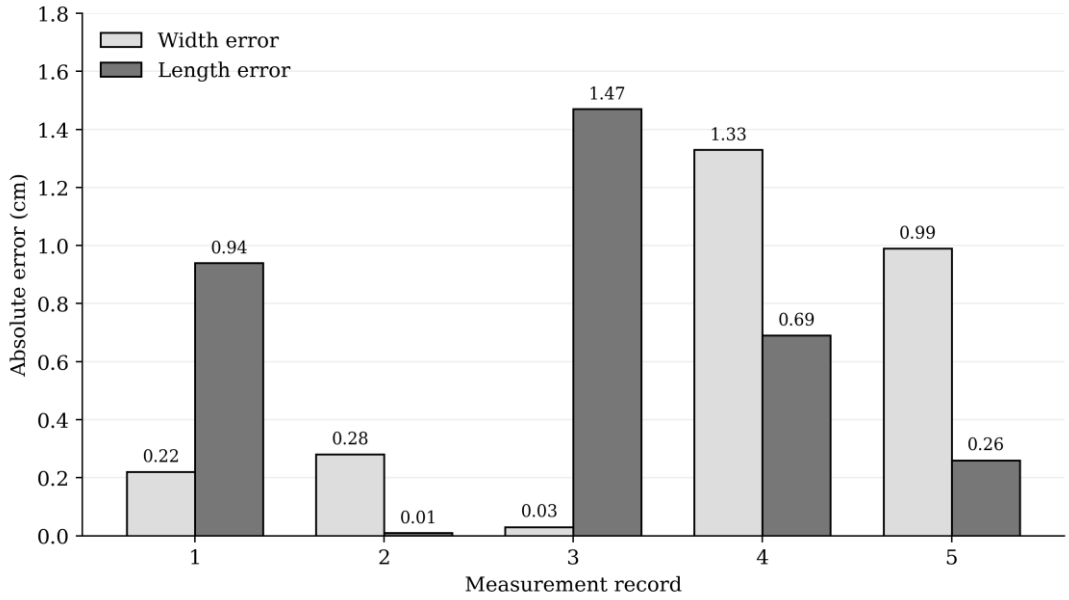


Figure 6.2: Absolute width and length errors across five measurement records

Source: absolute differences calculated from the measurements in Table 6.3.

6.2 Product and Technical Results

Previously reported frontend checks included type checking, 57 automated tests and successful compilation of the Electron targets. The present revision adds current runtime-health evidence but does not independently rerun those earlier checks.

The supplied Size health response confirms healthy status and model loading on CPU. Earlier endpoint and negative-control checks support selected functional behaviours, subject to the evidence boundaries in Table 5.1.

The supplied Windows screenshots show the Electron launch output and a connected-backend indicator in the dashboard. Together, these observations support development-mode startup and selected software behaviours. They do not establish physical measurement accuracy, measured processing latency, or packaged-installer reliability.

6.3 User or Stakeholder Validation

Collecting footage from garment manufacturing environments is not equivalent to conducting user validation; that requires structured feedback provided by operators or stakeholders based on their use of the component.

As interview records, questionnaire responses, or usability task results have not yet been provided, it is not possible to state a user satisfaction percentage or indicate stakeholder approval.

During validation, tasks such as selecting the garment type, preparing an order, understanding measurement status, and retrieving records can be tested.

6.4 Integration Results

The provided workstation-selection screenshot displays all four research components within the shared desktop interface. The initial Windows output confirms the launch of the Electron application, while the sizing dashboard indicates the status of the connected backend.

Independent API verification also validated selected operations of the sizing service. This evidence supports the cross-platform approach and backend accessibility. However, the display of the four workstations does not establish successful end-to-end

functionality for the camera, model, or Internet of Things (IoT) workflows of each individual component.

The current integration maintains separate component services and databases. There is no automated transfer of garment-type predictions from the style/color service to the sizing service; the measurement workflow relies on operator-selected garment types.

6.5 Summary of Findings

The evidence now includes the training configuration, 50-epoch validation log, user-confirmed dataset counts, current Size runtime health and five tape-reference comparisons. Together these support model development, deployment configuration and a preliminary record-level dimensional evaluation.

Evaluation remains partial. Independent held-out test performance, broader dimensional accuracy, temporal improvement, reference size agreement, count reliability, latency and user acceptance are not established. Table 6.5 distinguishes supported findings from outstanding validation.

Table 6.5: Summary of findings and remaining validation

Area	Supported finding	Remaining evidence
Model and API operation	The supplied model loaded successfully, and selected endpoint and input-validation checks passed.	Independent test-set evaluation and wider real-garment coverage
Measurement output	Five tape comparisons: width MAE 0.570 cm; length MAE 0.674 cm	Larger identified garment set, landmark documentation and uncertainty evaluation
Temporal confirmation	Consecutive-frame confirmation is implemented.	Repeated presentations and comparison with single-frame measurements
Counting	The blank-image test produced no count increment, and the interface displays counted entries.	Controlled presentation/removal trials to identify duplicate and missed counts.
Application integration	Screenshots show shared workstation selection, Electron startup, and a connected-backend indicator.	Complete camera and hardware workflow validation.
Processing performance	The backend processed requests during software verification.	Response-time measurements on the target hardware.
User acceptance	No structured stakeholder-validation results are available in the supplied evidence.	Documented operator tasks and feedback.

7 DISCUSSION

7.1 Interpretation of Findings

The model log supports segmentation performance on the configured validation partition. At epoch 50, mask mAP@0.50 is 0.96033 and mask mAP@0.50–0.95 is 0.95529. The five physical comparison records provide preliminary dimensional evidence, with width MAE 0.570 cm and length MAE 0.674 cm. Runtime health confirms loading and configuration, not physical accuracy.

The pooled dimensional summary contains repeated observations of one garment, and the remaining garment identities and exact landmarks are incompletely documented. The largest recorded length error is 1.47 cm and width error is 1.33 cm. These observations identify a basis for further trials but do not establish population accuracy, size agreement or counting reliability.

7.2 Comparison with Literature Baselines and Existing Solutions

Prior garment-measurement studies support contour interpretation [3], segmentation and measurement-point extraction [1], and depth-based physical-distance estimation [2]. The present design follows these established principles while adding a project-specific temporal and record-management workflow.

Serrat et al. report MAE for jackets and pants [3], whereas Kim et al. report relative errors under a LiDAR-based setup [2]. The current five-record cm errors cannot establish superiority because garment categories, landmarks, sample sizes and acquisition conditions differ.

The comparison concerns design choices rather than demonstrated performance superiority. No matched evaluation against these systems or a single-frame baseline is available. Differences in garments, datasets, and experimental conditions prevent direct accuracy or speed rankings.

7.3 Research Contribution

The research contribution comprises the preparation of garment segmentation data from real-world footage and the application of a trained model within a controlled

measurement workflow. Dataset development included frame selection, polygon annotation, background-only images, and dataset versioning.

The component combines segmentation-based geometry with suitability checking, calibrated dimension conversion, temporal confirmation, and removal-dependent counting. Its contribution lies in adapting and integrating these methods for the stated operating context rather than introducing a new segmentation algorithm.

The implementation provides an artefact through which these design choices can be investigated. Claims concerning improved accuracy, stability, or counting reliability require the corresponding quantitative evaluation.

7.4 Product Contribution

The product contribution is an operator-facing garment-measurement component integrated into the shared desktop application. It provides garment-type selection, measurement feedback, chart-based size estimates, counts, and access to historical and order-related records.

Separate backend services and databases preserve component boundaries within the complete solution. The shared interface provides access to the four research features, while the Size component retains its own measurement responsibilities.

This implementation creates a basis for practical workflow evaluation. Reduced manual effort, increased throughput, and improved production efficiency remain potential benefits rather than demonstrated outcomes.

7.5 Limitations and Threats to Validity

The principal physical-evaluation limitation is the small five-record comparison, including two trials of the same garment. Exact measurement landmarks, tape uncertainty, failure presentations and individual camera conditions were not fully documented. The shirts retain the width/length order explicitly confirmed by the researcher; their geometry requires better landmark records before broad comparison.

Measurement validity depends on consistent endpoint definitions, camera geometry, image scale, and garment placement. Folding, occlusion, lighting changes, and

departures from the calibrated measurement plane may affect the outputs. Dataset coverage also limits generalization to unfamiliar garments and environments.

Related frames appearing across partitions could inflate model metrics; source-video and garment-level split independence has not been verified. The model log provides validation rather than independent test results. Physical records cover T-shirts and shirts only, so equivalent trouser performance is not established.

Finally, the absence of structured stakeholder feedback, measured latency, and complete hardware trials limits conclusions concerning usability and operational suitability. Software tests and model confidence values cannot substitute for these evaluations.

7.6 Implications

The findings provide a basis for expanding the preliminary evaluation. Priority investigations include an identified garment set, matched single-frame and temporal-confirmation trials, garment-removal scenarios, independent test-set validation and processing-time measurements on the intended hardware.

Practical operation requires clear guidance on camera configuration, garment placement, and the interpretation of size estimates. Calibration records and traceable model versions are necessary to identify changes affecting measurement behavior.

Broader evaluation across garment categories, manufacturing environments, and operators is needed before recommending routine production use. Such evidence would clarify the component’s operating limits and determine whether its potential workflow benefits are realized.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research developed a computer vision component for garment dimension measurement and size estimation under a controlled overhead camera arrangement. The component combines garment segmentation, suitability checks, calibrated dimension extraction, consecutive-frame confirmation, and repeated-count control.

The supplied training run contains 50 epochs of validation results. The final logged mask mAP@0.50 and mAP@0.50–0.95 are 0.96033 and 0.95529. Five tape-reference comparison records yield width MAE 0.570 cm and length MAE 0.674 cm. These preliminary results are limited by the small record set and incomplete landmark documentation, and do not establish counting reliability or general manufacturing suitability.

The work provides an implemented prototype integrated into the shared desktop application. Further quantitative and stakeholder evaluation is required to establish its performance and suitability for routine manufacturing use.

8.2 Achievement of Objectives

The development objectives were addressed through garment-footage collection, frame selection, polygon annotation, segmentation-model development, and implementation of the measurement workflow. The component also includes size-chart mapping, counting control, and record-management functionality.

The available evidence supports dataset partition counts, logged segmentation validation, current model loading and preliminary tape-reference dimensional errors. Evaluation objectives remain partially addressed: independent test performance, temporal improvement, reference size agreement, counting reliability, latency and usability require additional studies.

Objective achievement is therefore distinguished between implemented functionality and experimentally validated performance.

Table 8.1 summarizes objective achievement based on the available implementation and evaluation evidence.

Table 8.1: Objective achievement and evidence

Objective area	Status supported by available evidence
Garment data collection and dataset preparation	Dataset configuration supplied; 1,295 images with 1,070/141/84 split counts reported by researcher
Segmentation-model development	50-epoch validation log supplied; deployed run identity confirmed; independent test evaluation outstanding
Suitability and temporal-confirmation logic	Implemented; effectiveness requires controlled trials
Calibrated dimension measurement	Implemented; five-record tape comparison yields preliminary dimensional errors
Size estimation and interface	Size estimates displayed; reference-based agreement requires evaluation
Counting and record management	Implemented; history displayed and negative-control behavior verified
Performance evaluation	Partially evaluated; larger garment trials, independent test validation, latency and stakeholder evaluation outstanding

8.3 Individual Research and Product Contributions

The individual research contribution comprises garment-data preparation, suitable-frame selection, polygon annotation, and segmentation-model development. It also includes the application of segmentation outputs to garment-specific measurement geometry within a calibrated camera arrangement.

The product contribution is the garment-measurement component, including operator-selected garment types, measurement feedback, size estimates, counting control, and historical records. Its integration provides access to these functions through the shared desktop application.

These contributions relate to the individual measurement component. The models and algorithms belonging to the other research components are excluded from the individual contribution claim

8.4 Future Directions

The immediate priority is quantitative evaluation using garment-specific manual measurements and repeated presentations. This should establish dimensional error,

measurement variation, size agreement, and duplicate or missed counts. Segmentation should also be evaluated using held-out annotated images.

Further testing should examine different garment categories, backgrounds, lighting conditions, and placement variations. Camera and image-scale changes require calibration checks. Processing latency should be measured on the intended hardware, while structured operator tasks should assess usability.

Interface improvements include garment-specific measurement labels, clearer confirmation states, and consistency between overlays and stored measurements. Packaged installation, restart behaviour, and record persistence also require target-machine validation.

Automatic garment-type input from the style/colour component may be investigated as a future integration extension. Such functionality would require explicit communication interfaces and independent validation before replacing operator selection.

8.5 Commercialisation and Operational Adoption

The intended users are apparel-packing and quality-control operators who need garment dimensions, chart-based size estimates and traceable records. A potential offering would combine the desktop application, camera setup, calibration guidance and support. This is a proposed adoption route; no customer demand study or commercial deployment is claimed.

Operational adoption requires a controlled pilot using predefined tolerances, labelled garments, recorded failures and operator feedback. Hardware and setup costs should be obtained from actual quotations, while throughput and labour effects should be measured against the existing workflow. No cost-saving percentage, revenue projection or production-efficiency gain is asserted from the present evidence.

REFERENCES

[1] A. Paulauskaite-Taraseviciene et al., “An intelligent solution for automatic garment measurement using image recognition technologies,” Applied Sciences, vol. 12, no. 9, Art. no. 4470, 2022, doi: 10.3390/app12094470.

[2] S. Kim, H. Moon, J. Oh, Y. Lee, H. Kwon, and S. Kim, “Automatic measurements of garment sizes using computer vision deep learning models and point cloud data,” Applied Sciences, vol. 12, no. 10, Art. no. 5286, 2022, doi: 10.3390/app12105286.

[3] J. Serrat, F. Lumbreras, and I. Ruiz, “Learning to measure for preshipment garment sizing,” Measurement, vol. 130, pp. 327–339, 2018, doi: 10.1016/j.measurement.2018.08.019.

[4] K. He, G. Gkioxari, P. Dollár, and R. Girshick, “Mask R-CNN,” arXiv:1703.06870, 2017. Available: https://arxiv.org/abs/1703.06870.

[5] Ultralytics, “Instance segmentation with Ultralytics YOLO,” Ultralytics Documentation. Available: https://docs.ultralytics.com/tasks/segment/. Accessed: Oct. 11, 2026.

[6] Ultralytics, “Model training with Ultralytics YOLO,” Ultralytics Documentation. Available: https://docs.ultralytics.com/modes/train/. Accessed: Oct. 11, 2026.

[7] Ultralytics, “Model validation with Ultralytics YOLO,” Ultralytics Documentation. Available: https://docs.ultralytics.com/modes/val/. Accessed: Oct. 11, 2026.

[8] OpenCV, “Camera calibration and 3D reconstruction,” OpenCV Documentation. Available: https://docs.opencv.org/4.x/d9/d0c/group__calib3d.html. Accessed: Oct. 11, 2026.

[9] Z. Zhang, “A flexible new technique for camera calibration,” IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 22, no. 11, pp. 1330–1334, 2000, doi: 10.1109/34.888718.

[10] Roboflow, “Dataset versions,” Roboflow Documentation. Available: https://docs.roboflow.com/datasets/versions/dataset-versions. Accessed: Oct. 11, 2026.

[11] FastAPI, “FastAPI,” Official Documentation. Available: https://fastapi.tiangolo.com/. Accessed: Oct. 11, 2026.

[12] SQLite, “Appropriate uses for SQLite,” SQLite Documentation. Available: https://www.sqlite.org/whentouse.html. Accessed: Oct. 11, 2026.

[13] Electron, “Process model,” Electron Documentation. Available: https://www.electronjs.org/docs/latest/tutorial/process-model. Accessed: Oct. 11, 2026.

[14] Size Stream, “3D body scanning and measurement technology,” Official Website. Available: https://www.sizestream.com/. Accessed: Oct. 11, 2026.

GLOSSARY

Term	Definition for the report
Calibration	Establishing the relationship between image measurements and physical dimensions under specific camera conditions.
Detection zone	A specifically designated area within an image that includes views of the garments for the purpose of evaluating their measurements.
Garment mask	A pixel-level representation of the garment region produced by segmentation.
Instance segmentation	Identifying the pixels belonging to each individual object in an image.
Marker-free measurement	Measurement without a reference marker in each operational frame, using previously established calibration.
Measurement confirmation	Acceptance of a measurement after the component's suitability and consecutive-frame conditions are satisfied.
Model confidence	A model-generated score indicating confidence in a prediction; it does not directly represent dimensional accuracy.
Occlusion	The partial or complete obstruction of a garment by another object.
Pixels per centimeter	The number of pixels corresponding to one centimeter on the measurement surface.
Rearming	Preparing for the next garment once the removal of the current garment is confirmed.
Size estimation	Determining the clothing size by comparing the measured dimensions with a pre-determined size chart.
Temporal stability	Consistency of measurements across consecutive observations over time.

APPENDICES

This appendix presents supplementary evidence of dataset preparation, runtime configuration and the garment-measurement interfaces. Figure A.1 shows segmentation examples, Figure A.2 documents the active health response, and Figures A.3 and A.4 show stored history and analytics. Tables A.1 and A.2 provide artefact provenance and tape-reference measurement records.

Appendix A Research Instruments or Extended Results

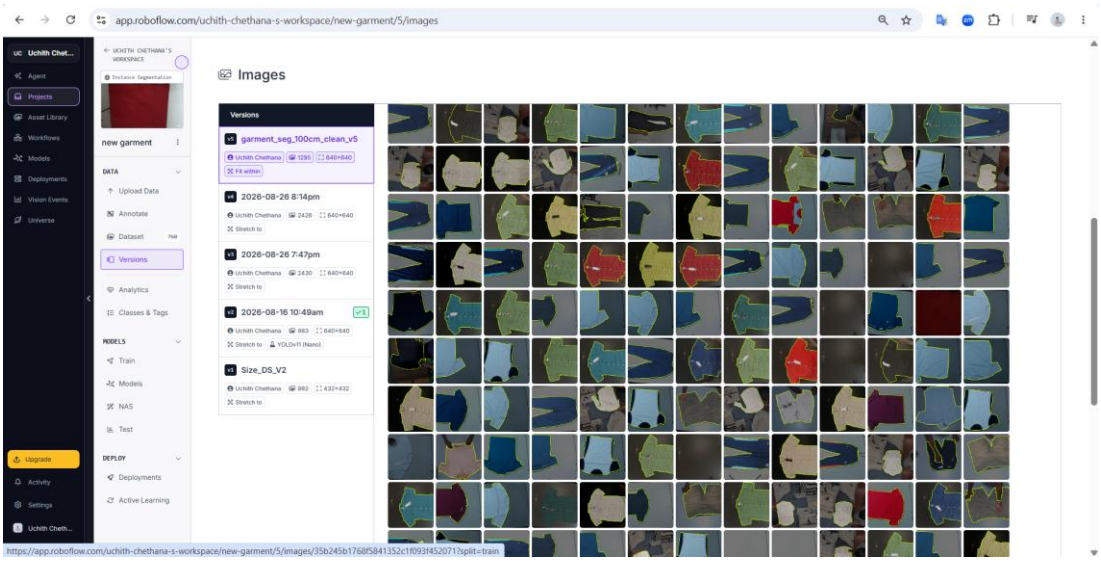


Figure A.1: Examples of garment images and segmentation annotations in the Roboflow dataset.

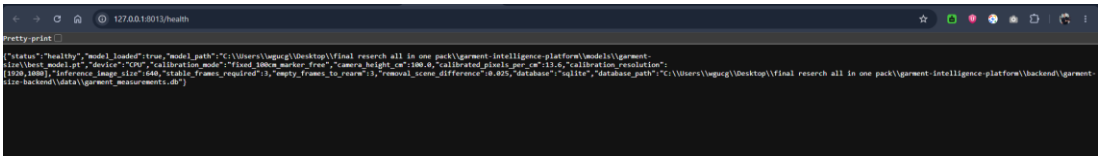


Figure A.2: Size-service health response and runtime configuration

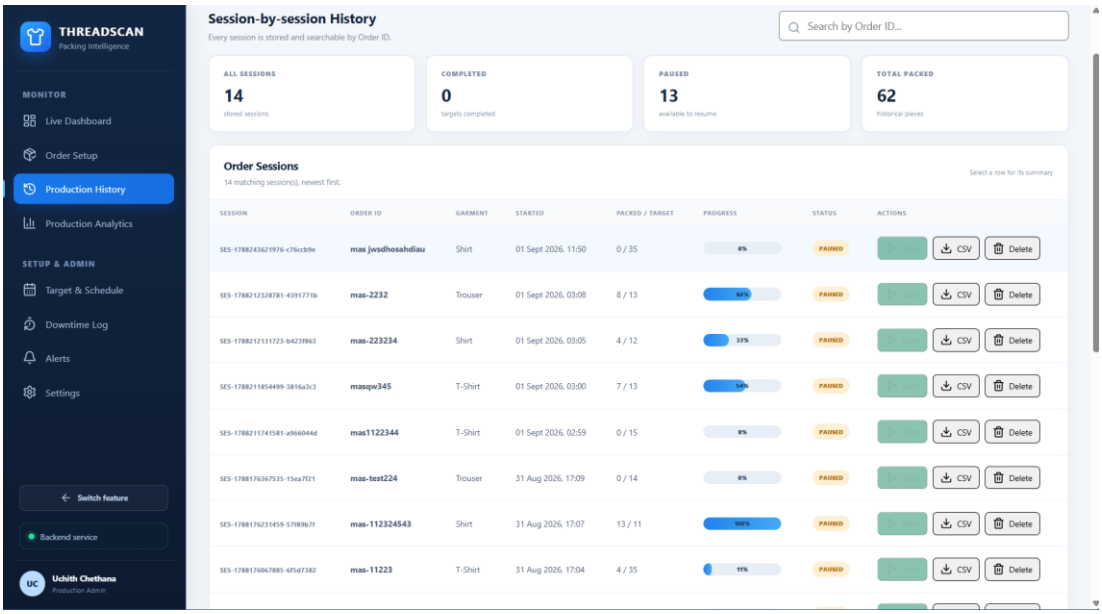


Figure A.3: Production-history interface showing stored sessions and recorded garment counts

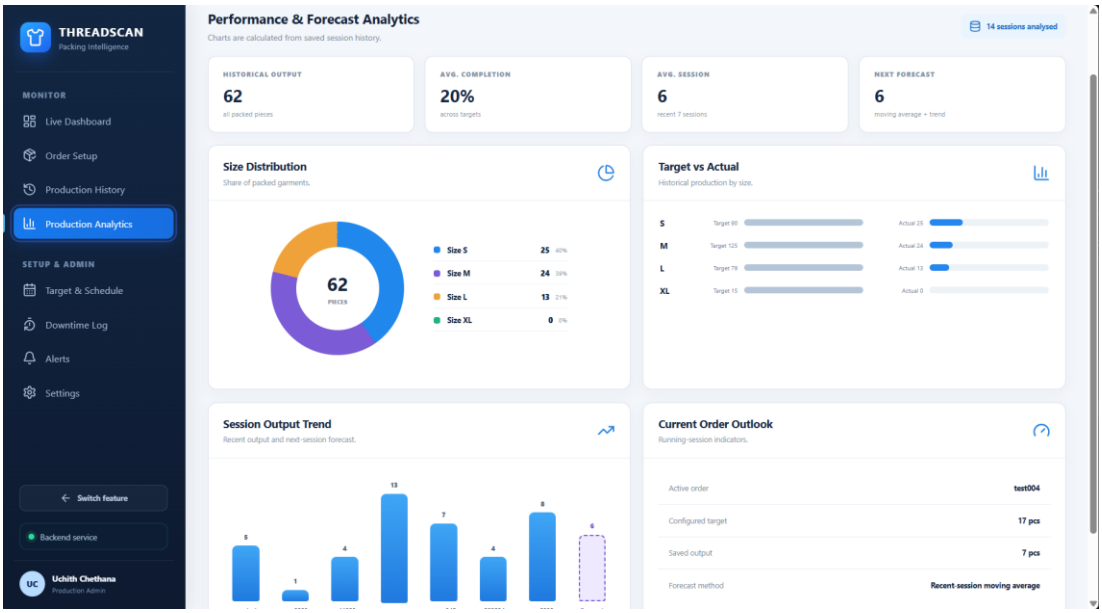


Figure A.4: Analytics interface summarising recorded session output and size distribution

A.1 Supplementary evidence index

Table A.1: Research artefacts and provenance

Artefact	Evidence used	Boundary
data.yaml	One class; version 5; configured split paths	Does not contain image counts
Dataset counts	Researcher supplied 1,070/141/84 images	Counts not independently recounted from dataset bytes
test01/args.yaml	Training settings in Table 3.2	Actual GPU model not recorded
test01/results.csv	50-epoch validation log	No independent test-set results
Runtime health screenshot	CPU; fixed calibration; image size 640	Run identity confirmed by researcher
Handwritten measurement sheet	Five tape and system records	Two trials share one garment; landmarks incomplete
Box and mask PR/F1 plots	Files supplied in training ZIP	Paired files are byte-identical; not treated as independent diagnostics

Table A.2 presents the tape-reference measurements and corresponding system outputs used in Tables 6.3 and 6.4. The values were transcribed from the original measurement sheet and confirmed by the researcher.

Table A.2: Tape-reference and system measurements of garments

Garment type	Tape width (cm)	Tape length (cm)	System width (cm)	System length (cm)	Estimated size
T-shirt	45.72	62.04	45.50	61.10	S
T-shirt	45.72	62.04	46.00	62.05	S
T-shirt	49.53	68.50	49.50	67.03	M
Shirt	68.58	45.72	69.91	45.03	S
Shirt	71.12	48.26	72.11	48.00	M

Note: Records 1 and 2 are repeated trials of the same garment. All dimensions are expressed in centimetres. The recorded width and length values were retained as confirmed by the researcher. Sizes are system estimates; independent reference-size labels were not supplied.

Appendix B Academic integrity and AI use disclosure

Table B.1: Academic integrity and AI use disclosure

AI tool used	Purpose	Representative prompts	Verification performed	Selection or artifacts affected
OpenAI ChatGPT	Used as a planning aid to organize the thesis structure	Requested suggestions for organizing chapters	Suggestions were independently reviewed against the module guidelines, project scope and available evidence.	Planning notes, chapter organization and evidence placement plan
	Prepare checklists	Planning the research workflow		
	Identify suitable locations for diagrams and screenshots	Positioning technical evidence within the report		